

LIST OF PUBLICATIONS

Marco Caponigro

- [1] *Controllability on the Group of Diffeomorphisms*,
Annales de l'Institut Henri Poincaré (C) Analyse Non Linéaire, 26 (2009), pp. 2503-2509.

Abstract

Given a compact manifold M , we prove that any bracket generating family of vector fields on M , which is invariant under multiplication by smooth functions, generates the connected component of the identity of the group of diffeomorphisms of M .

- [2] *Dynamics Control by a Time-varying Feedback*,
Journal of Dynamical and Control Systems, Vol. 16 (2010), 2 (April), pp. 149–162.

Abstract

We consider a smooth bracket generating control-affine system in $\mathbb{R}^n$ and show that any orientation preserving diffeomorphism of $\mathbb{R}^n$ can be approximated, in the very strong sense, by a diffeomorphism included in the flow generated by a polynomial with respect to state variables and trigonometric polynomial with respect to time variable time-varying feedback control.

- [3] *Families of vector fields which generate the group of diffeomorphisms*, arXiv:0804.4403. (2008)
to appear on Proc. Steklov Math. Inst.

Abstract

Given a compact manifold M , we prove that any bracket generating family of vector fields on M , which is invariant under multiplication by smooth functions, generates the connected component of the identity of the group of diffeomorphisms of M .

Notes:

- (a) [1] and [2] in joint work with Andrei A. Agrachev.
- (b) In [1] and [3] the same result has been proved. In [3] we used Nash–Moser Implicit Function Theorem, while in [1] we give a simpler proof using method of characteristics.